**SKEF2123 INSTRUMENTATION & MEASUREMENT**

Section EB1

**TECHNICAL REPORT (10%)**

# Electrical Measurement


Group 9

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# 1.0 INTRODUCTION

An ammeter is a device used to measure the electric current flowing through a circuit. An ammeter always must be connected in series with other component so that the current passes directly through the meter because it measures the flow of electrons. An ammeter ideally should have zero internal resistance so that it does not block the current that it is trying to measure.

A voltmeter is a device used to measure the potential difference or voltage between two points. A voltmeter is connected in parallel across the component being measured. This will allow it to measure the voltage without interrupting the current flow. A voltmeter is designed to have very high resistance. Both ammeter and voltmeter rely on current to produce a reading although they measure different things.

An analog instrument uses a needle or pointer that moves continuously across a scale to show the measurement. A digital instrument displays the reading directly as digit numbers on a screen. Digital meters are commonly used today because they are easier to read and provide higher accuracy without errors that come from reading a moving needle.

Wheatstone's bridge is used to measure resistance precisely. The circuit is designed to measure an unknown electrical resistance by balancing two legs of a circuit. It works like a balanced scale, we adjust the known variable resistor until the current flowing through the centre meter is zero. The state is called balanced. When the bridge is balanced, we can calculate the unknown resistance.

# 2.0 OBJECTIVES

1. To perform current and voltage measurements using electrical and electronic instruments.
2. To perform resistance measurement using Wheatstone's bridge.

# 3.0 APPARATUS

1. Digital Multimeter
2. Analog Multimeter
3. Wires

4. Solderless board
5. Power supply / 9V Battery
6. Resistor: 100 Ω, 150 Ω, two 330 Ω, 1 kΩ and 39 kΩ
7. Potentiometer

# 4.0 PROCEDURES

## Part A: Voltage and Current Measurement

1. The circuit is connected by using the provided apparatus as shown in Figure 1.1.

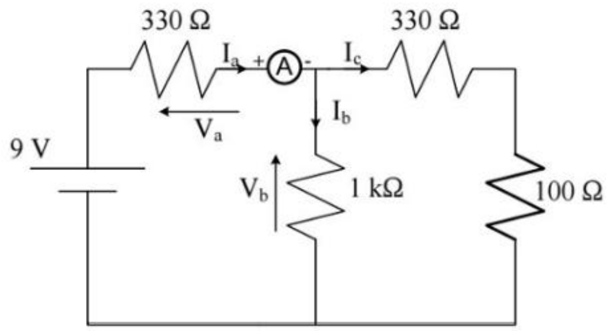



Figure 1: Circuit diagram

2. $I_a, I_b, I_c, V_a, V_b$ of Figure 1 are calculated by using Kirchhoff's circuit laws.
3. Appropriate electrical instruments are used to measure Ia, Ib, Ic, Va and Vb.
4. All the parameters are recorded in the table below.

| Parameter | Calculation | Measurement 1 (Analog Instruments) | Measurement 2 (Digital Instruments) |
|---|---|---|---|
| $I_a$(mA) | 14.27 | 14.0 | 13.65 |
| $I_b$(mA) | 4.29 | 4.5 | 4.54 |
| $I_c$(mA) | 9.98 | 9.5 | 9.24 |
| $V_a$(V) | 4.29 | 4.5 | 4.54 |
| $V_b$(V) | 4.71 | 4.4 | 4.43 |

5. The values of the obtained parameters are compared and concluded.
6. Experiences and observations while using the multimeters for electrical measurements are discussed.

## Part B: Resistance Measurement – Wheatstone's bridge

1. The circuit is connected as shown in Figure 2.

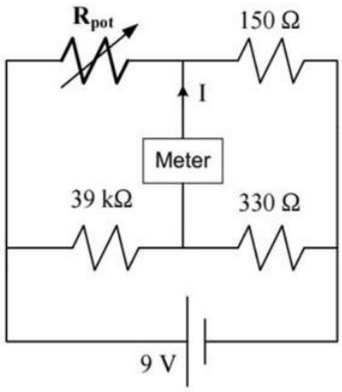



Figure 2: Wheatstone's Bridge

2. The theoretical value of the potentiometer ($R_{pot}$) required to balance the bridge is calculated.

3. The knob of the potentiometer is turned until the meter in the meter reads 0 V which is the balanced condition.

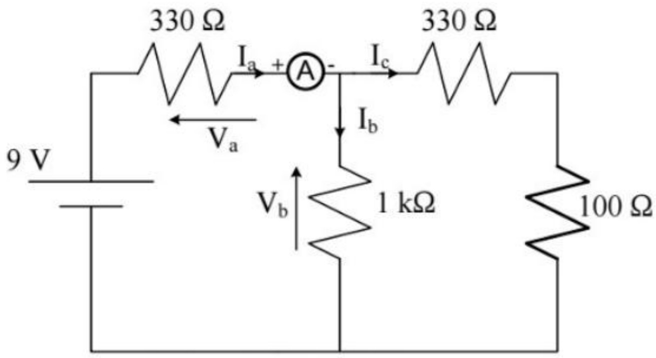
4. The potentiometer is disconnected and its resistance is measured using a multimeter. The reading is recorded as $R_{pot,balanced}$.

5. 10 different knob positions from scale 1 to 10 are selected and the voltage across the bridge for each are measured. A graph of voltage vs. resistance is plotted.

# 5.0 RESULT AND DISCUSSION

## Part A: Voltage and Current Measurement

Calculation (Working Step)




$$I_{entering} = I_{exiting}$$
$$I_a = I_b + I_c$$

$$V = IR$$
$$I = \frac{V}{R}$$
$$V_a = 9 - V_b$$

$$I_a = \frac{9 - V_b}{330} \qquad I_b = \frac{V_b}{1000} \qquad I_c = \frac{V_b}{330 + 100}$$
$$I_c = \frac{V_b}{430}$$

$$\frac{9 - V_b}{330} = \frac{V_b}{1000} + \frac{V_b}{430}$$

$$\frac{9}{330} = \frac{V_b}{330} + \frac{V_b}{1000} + \frac{V_b}{430}$$

$$\frac{9}{330} = V_b\left(\frac{1}{330} + \frac{1}{1000} + \frac{1}{430}\right)$$

$$\boxed{\mathbf{V_b = 4.2909\ V}}$$

$$V_a = 9 - V_b$$
$$V_a = 9 - 4.2909$$
$$\boxed{\mathbf{V_a = 4.7091\ V}}$$

$$I_a = \frac{9 - V_b}{330}$$

$$I_a = \frac{9 - 4.2909}{330}$$

$$I_a = 0.01427\ A$$
$$\boxed{\mathbf{I_a = 14.27\ mA}}$$

$$I_b = \frac{V_b}{1000}$$

$$I_b = \frac{4.2909}{1000}$$

$$I_b = 0.00429\ A$$
$$\boxed{\mathbf{I_b = 4.29\ mA}}$$

$$I_c = \frac{V_b}{430}$$

$$I_c = \frac{4.2909}{430}$$

$$I_c = 0.00998\ A$$
$$\boxed{\mathbf{I_c = 9.98\ mA}}$$

## Measurement (Analog and Digital Instrument)

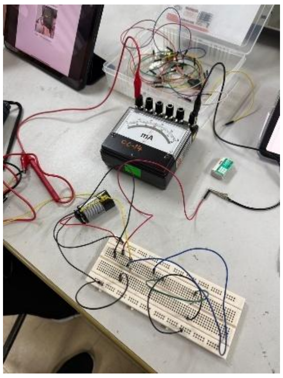


Figure 3.1 Overall circuit

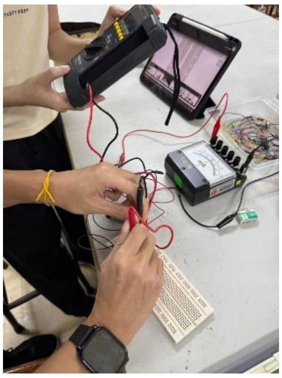


Figure 3.2 Zero error of multimeter

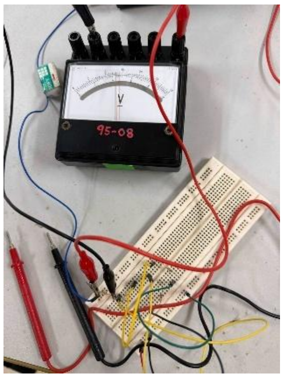



Figure 3.3 Circuit to measure $V_a$ with analog multimeter

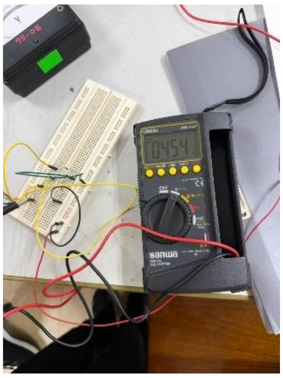



Figure 3.4 Circuit to measure $V_a$ with digital multimeter

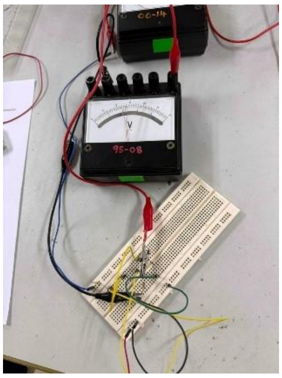



Figure 3.5 Circuit to measure $V_b$ with analog multimeter

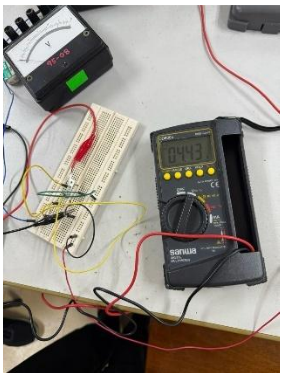



Figure 3.6 Circuit to measure $V_b$ with digital multimeter

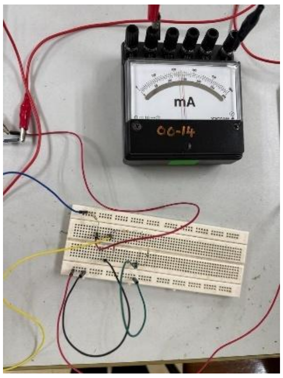



Figure 3.7 Circuit to measure $I_a$ with analog multimeter

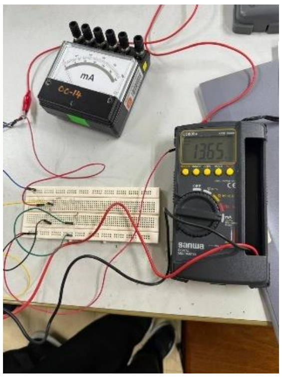



Figure 3.8 Circuit to measure $I_a$ with digital multimeter

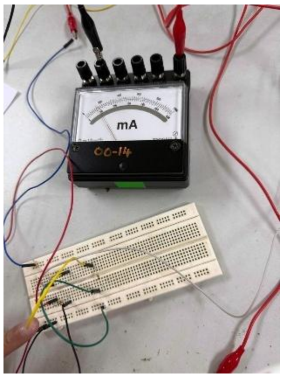



Figure 3.9 Circuit to measure $I_b$ with analog multimeter

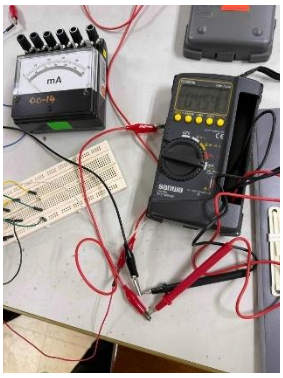


Figure 3.8 Circuit to measure $I_b$ with digital multimeter

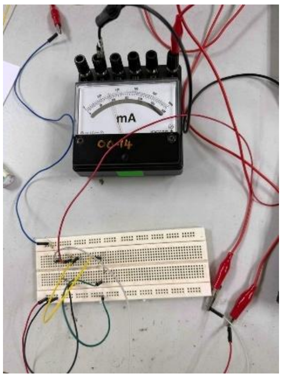



Figure 3.10 Circuit to measure $I_c$ with analog multimeter

Figure 3.11 Circuit to measure $I_c$ with digital multimeter

## Observation

1. We found that reading analog multimeter was harder because we had to look straight at the needle to avoid parallax error. Digital multimeter directly showed us the exact number on the screen, easier for us to get a precise reading.

2. When we added the voltage measured across the first resistor and the voltage across the second resistor, the total voltage was almost equal to the 9V from the battery. It showed that the total voltage supplied is shared between the components in the circuit.

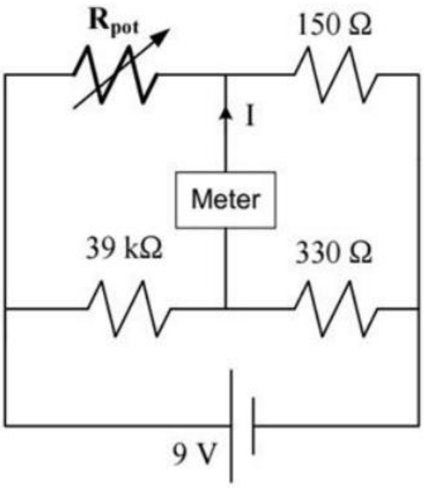
## Discussion

1. The actual measured values were lower than the calculated theoretical value. When we connect a voltmeter, it requires a small amount of current to function which will slightly lower the voltage. It is called loading effect. Also, adding an ammeter will also add a small amount of resistance to the circuit which will slightly reduce the total current.

2. We connected the ammeter in series because it has very low resistance and needs the current to flow through. We connected the voltmeter in parallel because it has very high resistance and should not disturb the current flow. If we connected it wrongly, it could have blown the fuse or gotten zero readings.

# Part B: Resistance Measurement – Wheatstone's bridge

Calculation (Working Step)




$$R_{pot} = \frac{R_1 \times R_2}{R_3}$$

$$R_{pot} = \frac{R_1 \times R_2}{R_3}$$

$$R_{pot} = \frac{150 \times (39 \times 10^3)}{330}$$

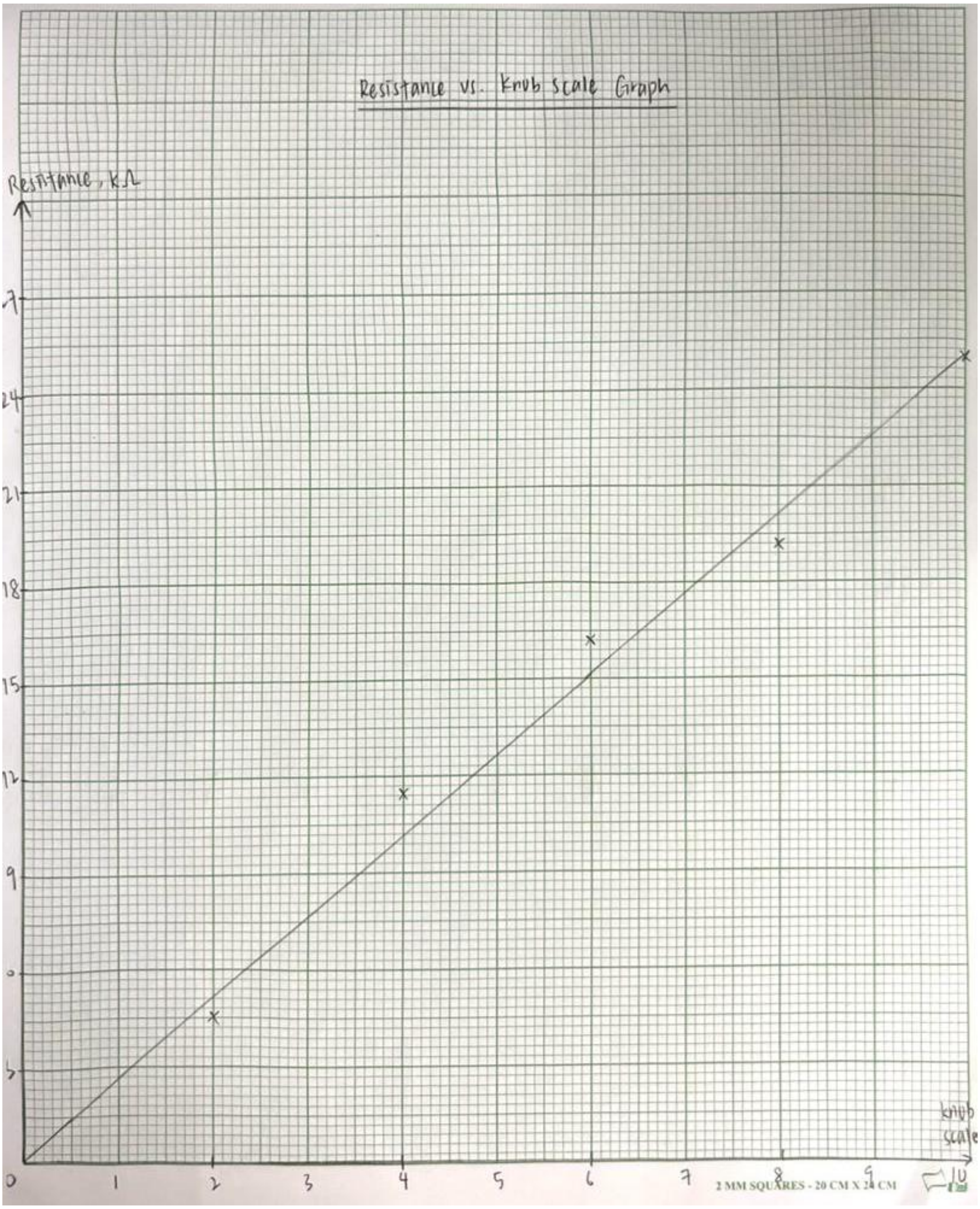
$$R_{pot} = 17727.27\ \Omega$$

$$R_{pot} \approx 17727\ \Omega$$

$$\boxed{R_{pot} \approx 17.73\ \text{k}\Omega}$$

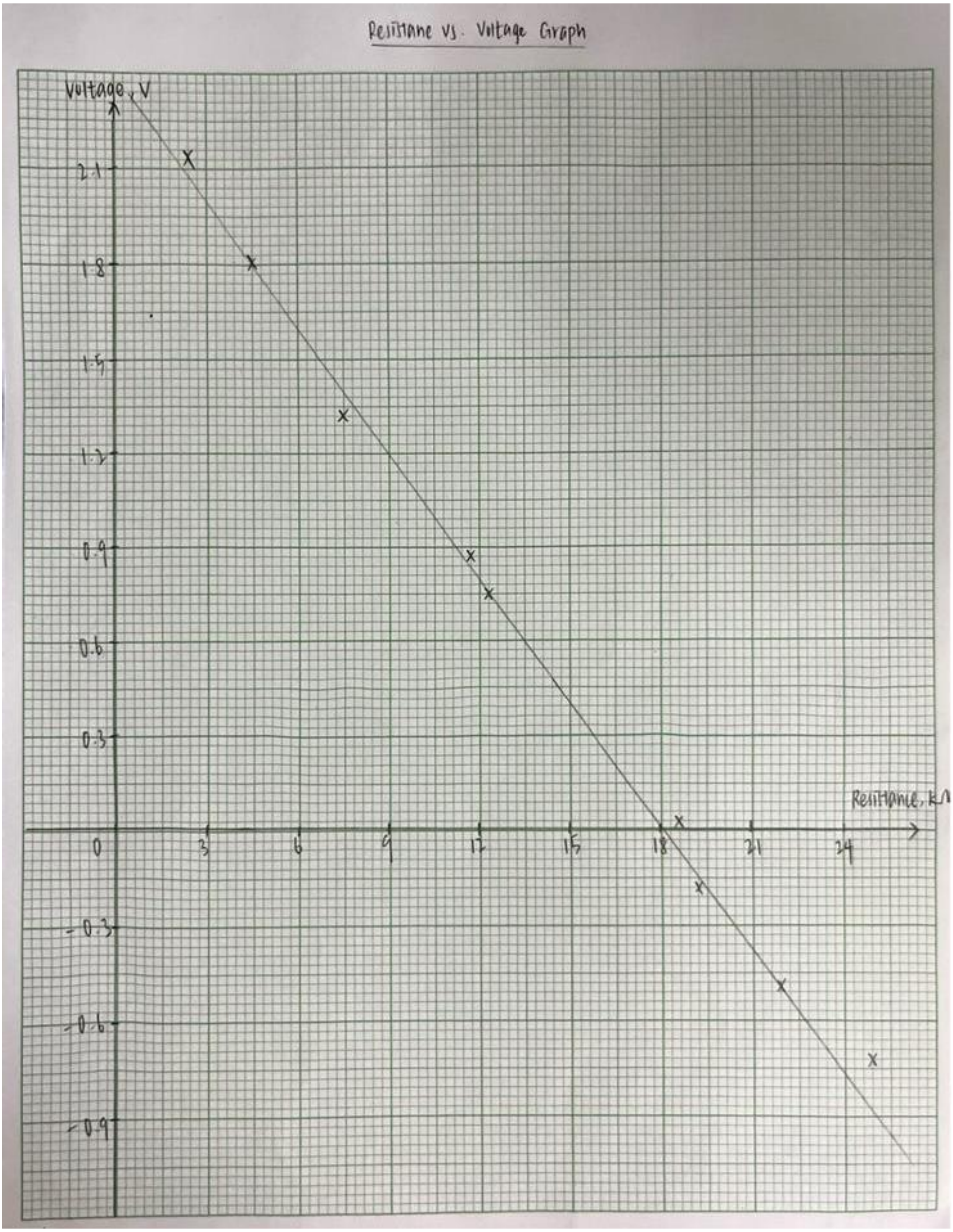
Measurement

| Scale | Resistance (kΩ) | Voltage (V) | Current (mA) |
|---|---|---|---|
| 1 | 2.52 | 2.25 | 2.21 |
| 2 | 4.53 | 1.88 | 1.85 |
| 3 | 7.49 | 1.39 | 1.33 |
| 4 | 11.58 | 0.87 | 0.86 |
| 5 | 12.55 | 0.75 | 0.73 |
| 6 | 16.18 | 0.25 | 0.24 |
| 7 | 18.54 | 0.02 | 0.02 |
| 8 | 19.43 | -0.21 | -0.20 |
| 9 | 22.48 | -0.51 | -0.49 |
| 10 | 25.21 | -0.73 | -0.71 |



Resistance vs. knob scale Graph
Resistance, kΩ
knob scale
0
1
2
3
4
5
6
7
8
9
10
9
12
15
18
21
24
2 MM SQUARES - 20 CM X 24 CM

| $R_{pot,balanced}$ (kΩ) | |
|---|---|
| **Calculation** | **Measurement** |
| 17.73 | 18.54 |



Observation

1. When we turned the knob of the potentiometer, the needle on the centre meter moved left and right. We found that at one specific position, the needle pointed exactly at zero. The state is called balanced condition.

2. We noticed that even a very small turn of the potentiometer knob will cause the metre needle to move far away from zero. This shows that the Wheatstone's bridge is very sensitive and can detect very small change in resistance which will allow us to get very accurate measurement.

Discussion

1. When the bridge is balanced, it means that the ratio of the resistors on the left side is the same to the ratio of the resistors on the right side. Because the current in the middle is zero, the readings do not depend on the accuracy of the meter but on the accuracy of the resistors used.

2. A standard ohmmeter uses battery and a spring mechanism that can weaken over time, and will cause errors. The Wheatstone's bridge uses the balanced condition, meaning we only look for the zero reading. This method can eliminate many errors.

# 6.0 CONCLUSION

In conclusion, this experiment successfully achieved the objectives of performing electrical measurements using different instruments and methods.

In part A, we successfully applied Ohm's Law and Kirchhoff's Law to analyse a DC circuit. We compared our calculated values with experimental results from analog and digital multimeters. We found that digital meters provide more precise readings.

In Part B, we use Wheatstone Bridge to find an unknown resistance value by balancing the circuit. The theoretical resistance value ($\mathrm{R} = 17.73\ \mathrm{k\Omega}$) was almost the same as the experiment measured value ($\mathrm{R} = 18.54\ \mathrm{k\Omega}$). There is a minor deviation caused by practical factors such as component tolerances and contact resistances. The experiment showed the high accuracy of bridge circuits for measuring resistance.